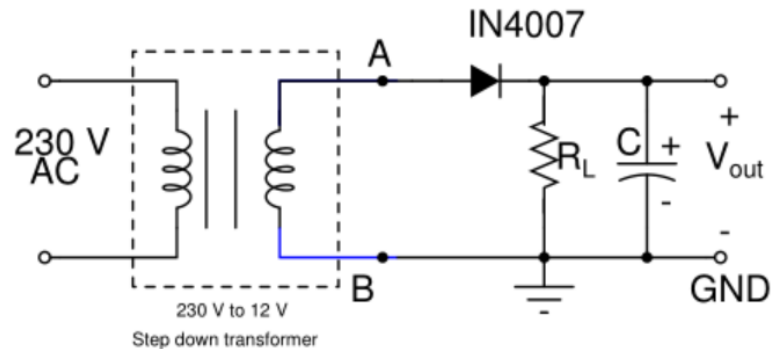


Sample Pre-Lab Quiz Questions for Expt 3

Duration: 10 min; Max marks: 5 marks

1. A rectifier circuit is shown below. A 230 V (RMS) to 12 V (RMS) step down transformer is used as the input to the rectifier circuit. Mark **all the correct options** with regard to the given circuit.

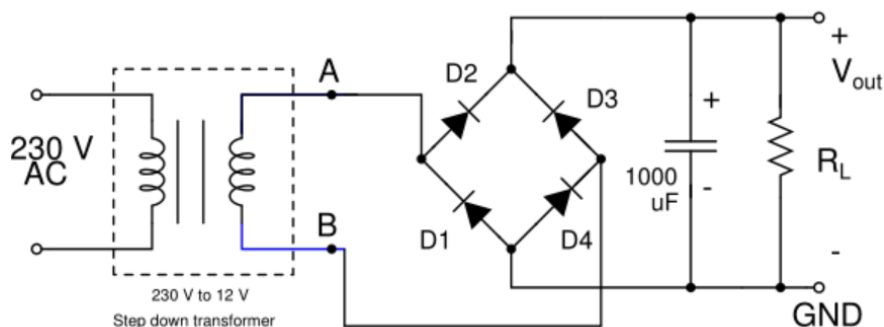


- ☒ A) The circuit is a half wave rectifier with capacitor filter.
- ☐ B) The circuit is a full wave rectifier with capacitor filter.
- ☒ C) The diode will conduct for the entire positive half cycle of the transformer secondary voltage.
- ☐ D) The diode will conduct for the entire negative half cycle of the transformer secondary voltage.
- ☐ E) With the capacitor C not present, V_{out} will be a flat (constant) DC voltage.
- ☒ F) With the capacitor C not present, V_{out} will be a unidirectional, but pulsating DC voltage.
- ☐ G) With the capacitor C not present, V_{out} will be an AC voltage.

2. With reference to the rectifier circuit shown above, mark **all the correct options**.

- ☒ A) For a given C (say $100\ \mu\text{F}$) if R_L is very large (say $1\ \text{M}\Omega$), V_{out} will be almost a flat DC voltage.
- ☐ B) For a given C (say $100\ \mu\text{F}$) if R_L is reduced, say from $10\ \text{k}\Omega$ to $500\ \Omega$, the ripple voltage in the V_{out} waveform will decrease.
- ☒ C) For a given C (say $100\ \mu\text{F}$) if R_L is reduced, say from $10\ \text{k}\Omega$ to $500\ \Omega$, the ripple voltage in the V_{out} waveform will increase.
- ☐ D) For a given R_L (say $1\ \text{k}\Omega$) if C is increased, say from $100\ \mu\text{F}$ to $470\ \mu\text{F}$, the peak-to-peak ripple voltage in the V_{out} waveform will increase.
- ☒ E) For a given R_L (say $1\ \text{k}\Omega$) if C is increased, say from $100\ \mu\text{F}$ to $470\ \mu\text{F}$, the peak-to-peak ripple voltage in the V_{out} waveform will decrease.

3. A rectifier circuit is shown below. A 230 V (RMS) to 12 V (RMS) step down transformer is used as the input to the rectifier circuit. Mark **all the correct options** with regard to the given circuit.



- A) The circuit is a half wave rectifier with capacitor filter.
 B) The circuit is a full wave rectifier with capacitor filter.
 C) During the positive half cycle of V_{AB} , diodes D3 and D1 will conduct.
 D) During the positive half cycle of V_{AB} , diodes D2 and D4 will conduct.
 E) For the same values of R_L and C , the peak-to-peak ripple voltage in the output voltages (V_{out}) of the circuit in Q.1 and this circuit will be the same.
 F) For the same values of R_L and C , the peak-to-peak ripple voltage in the output voltage (V_{out}) of the circuit in Q.1 will be more than that of this circuit.
 G) For the same values of R_L and C , the peak-to-peak ripple voltage in the output voltage (V_{out}) of the circuit in Q.1 will be less than that of this circuit.
5. With reference to the circuit shown in Q.3, mark **all the correct options**.
 A) The transformer secondary voltage (V_{AB}) should not be observed on the DSO by connecting a DSO probe across A and B.
 B) The transformer secondary voltage (V_{AB}) may be observed on the DSO by connecting a DSO probe across A and B.
 C) A DC power supply made using a half wave rectifier is called a regulated power supply because for a given transformer output voltage V_{AB} , the output voltage V_{out} remains constant for large range of R_L values.
 D) A DC power supply made using a full wave rectifier is called a regulated power supply because for a given transformer output voltage V_{AB} , the output voltage V_{out} remains constant for large range of R_L values.
 E) The DC power supply circuits shown in Q1 and Q3 are both unregulated power supplies because for a given transformer output voltage V_{AB} , the output voltage V_{out} will vary when R_L is varied.
6. Mark **all the correct options** with regard to the LM 7805 regulator DC power supply. The input to the regulator circuit is the output of the bridge rectifier circuit.
 A) For a large range of load currents (say up to 1A), the regulator will give a steady 5 V output.
 B) Only for a medium range of load currents (say up to 100 mA), the regulator will give a steady 5 V output.
 C) Only for a small range of load currents (say up to 20 mA), the regulator will give a steady 5 V output.
 D) For the regulator circuit to work properly, its input voltage should be between 12 V and 25 V.
 E) For the regulator circuit to work properly, its input voltage should be between 15 V and 25 V.
 F) For the regulator circuit to work properly, its input voltage should be between 7 V and 25 V.

---- The End ----